

Name: _____

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Class: _____

MyDesign Portfolio

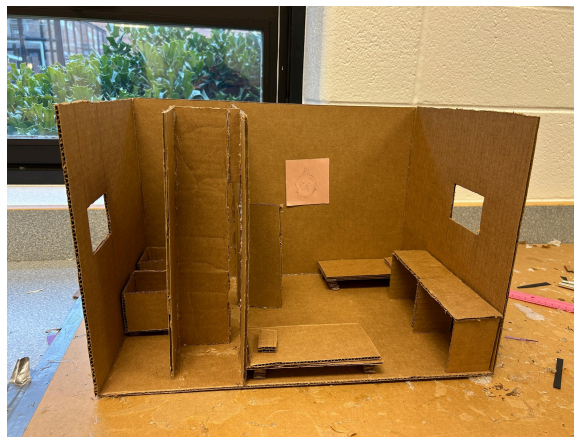
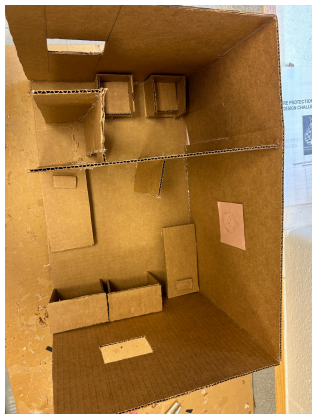
Element G: Construction of a Testable Prototype

Prototype Construction of Design #1

Date: March 14th, 2025

Photos of the fully built prototype:

This prototype was generally just a floor plan design of the model apartment. We placed everything in specific spots to ensure that our suppression system would be able to work and fit into the living space.



	Details	Tools and Manufacturing Methods used
Step 1	We were given materials to build our prototype.	Cardboard, hot glue, paper sticky note, handheld saw, scissors.
Step 2	We sketched out the dimensions of the walls and furniture,	Pencils, rulers, cardboard.

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	and mapped out windows and doors.	
Step 3	We cut out the walls, doors, windows and furniture for the prototype.	Cardboard, handheld saw, scissors, pencils.
Step 4	We glued all 4 walls and the floor together to create the bare floor plan.	Cardboard, hot glue.
Step 5	We assembled and designed the furniture together. (Beds, Chairs, Desks and a Poster)	Cardboard, hot glue, pencils, sticky notes.
Step 6	We positioned the furniture into the previously assembled apartment floor plan.	Previously assembled furniture
Step 7	Take photos of the fully assembled apartment prototype.	Phone camera

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #2

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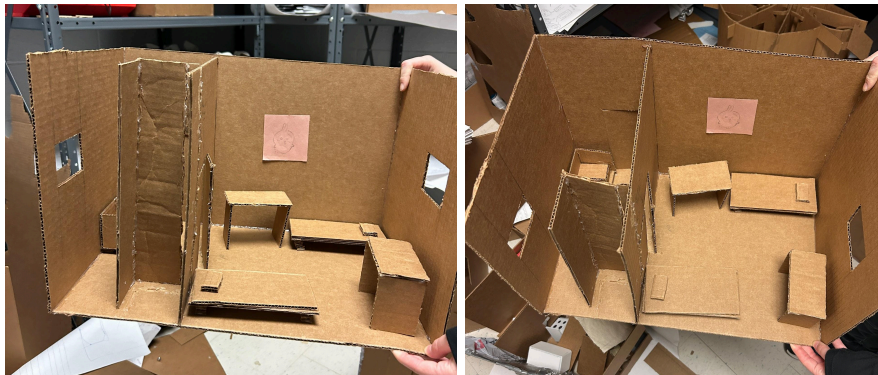
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Date: March 28th, 2025

Photos of the fully built prototype:

This prototype was used to size our suppression system based on the scenario of people actually living there. The plastic tubing would have ran across the ceiling and the water tank would have been put into the closet space in the living room area.



	Details	Tools and Manufacturing Methods used
Step 1	We had our previous prototype built but the door wasn't in the right spot, so we replaced the door and put it in a new spot.	Cardboard, hot glue, paper sticky notes, handheld saw, scissors.
Step 2	We placed the plastic tubing along the top of the walls/"ceiling".	Plastic PVC piping
Step 3	We placed the water pressure pump into the closet.	Water pressure pump

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Step 4	We removed the pump and began coding and building the electrical system.	Arduino kit, wires, light, sensor, buzzer
Step 5	We finalized the placement of the furniture.	Cardboard furniture
Step 6	We removed the plastic piping from the walls/"ceiling".	Plastic PVC piping
Step 7	We continued coding the suppression system.	Arduino kit

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

- We needed to make sure the suppression system worked.
- We needed to make sure the suppression system fit into the space and was livable.
- The suppression system didn't work the first few times we tried it so we kept coding until it was working again.

List of Materials:

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
Water Pressure Pump	N/A	Amazon.com	\$6.00	1	\$6.00
Plastic PVC Piping	6 feet long	Amazon.com	\$5.00	1	\$5.00

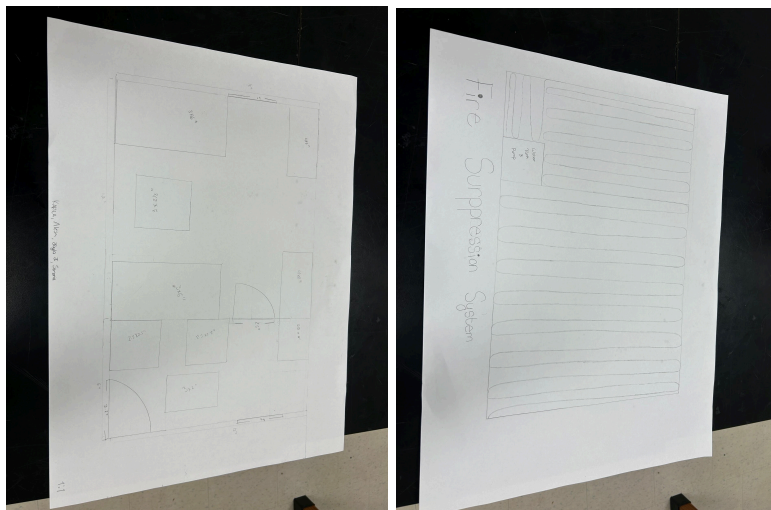
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Cardboard	N/A	N/A	N/A	5	N/A
Hot Glue	N/A	N/A	N/A	10	N/A
Sticky Note	N/A	N/A	N/A	1	N/A
Arduino Kit	N/A	N/A	N/A	1	N/A
Pencils	N/A	N/A	N/A	4	N/A

Detailed sketches, which might include CAD if you have learned it in your class:



Design Requirement	How will this prototype allow you to collect objective, measurable data?
Must pump enough water.	It will allow us to try until we know how much water we need until the fire is suppressed.
Must have windows and doors.	It will allow us to have a secondary way to suppress the fire and have an evacuation route.

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Apartment must be livable.	The space must be able to be lived in because there needs to be flammable materials to put the system to the test.
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Attribute	Description	Describe your prototype's functionality to test this attribute
Holes in the tubing	Our plastic tubing needed to have holes in them in order for the water to be released.	Expert Review Required
Water pump to pump water	The pump was needed to get the water from the tank into the tubing to suppress the fire.	Expert Review Required
Sensor connection to the pump	The sensor needed to connect to the pump to signal it to pump water.	Expert Review Required

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Holes in tubing	The ceiling is detachable, it might create stability issues among the tubing.
Water pump to pump water	Not having enough water to suppress the fire all the way out.
Sensor connection to the pump	Something may go wrong with the connection and coding that could prevent the system from working passed the sensory point.



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